Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 4 — Graphs and transformations Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第4章 · 图像与变换
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C04-R01 · P1 · Chapter 4

The curve C has equation $y = (x + 2)(x - 1)(x - 4)$.

(a) Sketch C, showing the coordinates of all intersections with the coordinate axes. (3)

(b) The curve is translated by $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Find the coordinates of the image of $(0, 8)$ and write the equation of the translated curve. (3)

(Total for Question 1 is 6 marks)

The graph of $y = f(x)$ passes through $A = (-3, 2)$, $B = (1, -4)$ and $C = (4, 1)$.

(a) For the graph $y = 2f(x - 1) + 3$, find the images of

(i) A, (2)

(ii) B and C. (2)

(b) Describe a sequence of transformations taking $y = f(x)$ to $y = 2f(x - 1) + 3$. (3)

(Total for Question 2 is 7 marks)

The curve C is $y = \dfrac{4}{x}$.

(a) Sketch C, showing its asymptotes and the points on C with $x = \pm 2$. (3)

(b) The curve is translated 3 units left and 1 unit down. Find

(i) the equation of the new curve, (2)

(ii) its intersections with the coordinate axes. (2)

(Total for Question 3 is 7 marks)

The graph of $y = f(x)$ is shown in Figure 4. The marked points are $(-4, 0)$, $(-1, 3)$, $(2, -2)$ and $(5, 0)$; straight segments join consecutive points.

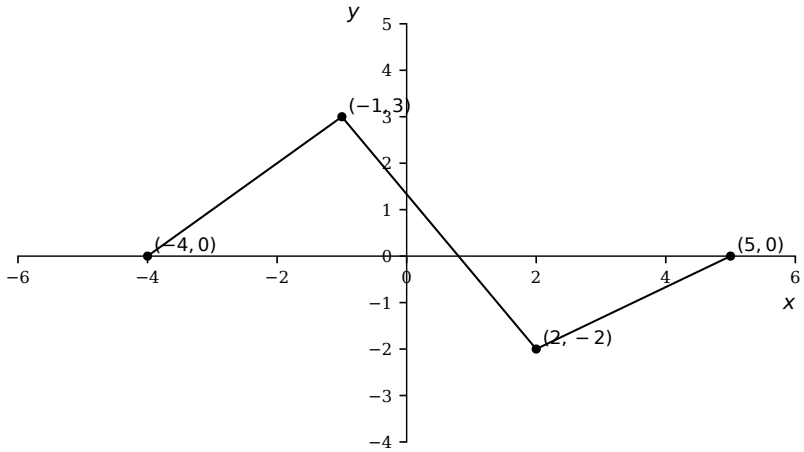


Figure 4

(a) Find the coordinates of the images of all four marked points under the transformation $y = -f(x)$. (2)

(b) Sketch $y = f(-x)$, labelling the images of the four marked points. (3)

(Total for Question 4 is 5 marks)

C is the curve $y = (x - 2)^2 + 1$.

(a) State the minimum point of C and explain why C does not intersect the x-axis. (3)

(b) C is stretched parallel to the y-axis by factor 3, then translated down 5. Find

(i) the equation and minimum point of the resulting curve, (3)

(ii) the exact x-intercepts. (2)

(Total for Question 5 is 8 marks)

The curve C is $y = \dfrac{6}{x^2}$.

(a) Sketch C, showing both asymptotes and the points at $x = \pm 1$. (3)

(b) After reflection in the x-axis and translation up 2, find the equation of the curve and its x-intercepts. (3)

(Total for Question 6 is 6 marks)

Point $P = (6, -3)$ lies on $y = f(x)$.

(a) Find the corresponding point on each graph:

(i) $y = f(2x)$, (2)

(ii) $y = \frac{1}{2}f(x)$. (2)

(b) Point $Q = (a, b)$ lies on $y = f(x)$. Give the corresponding point on $y = \frac{1}{2}f(2x)$. (2)

(Total for Question 7 is 6 marks)

8. P1-C04-R08 · P1 · Chapter 4

The curve C has equation $y = x^3$.

(a) Write the equation of the curve obtained by translating C so that $(0, 0)$ moves to $(2, -5)$. (2)

(b) Find the exact coordinates where the translated curve intersects the two coordinate axes. (3)

(Total for Question 8 is 5 marks)

9. P1-C04-R09 · P1 · Chapter 4

The graph of $y = f(x)$ has exactly three x-intercepts, at $x = -2, 1, 5$, and $f(0) = 4$.

(a) Find all x-intercepts of $y = f(3x)$. (2)

(b) For $y = f(x - 4)$, find all x-intercepts and the point corresponding to $(0, 4)$. (3)

(Total for Question 9 is 5 marks)

Curve A is $y = \dfrac{1}{x-2} + 3$ and curve B is $y = \dfrac{1}{x+1} - 2$.

(a) State the asymptotes of each curve. (2)

(b) Describe the single translation that maps A onto B. (2)

(Total for Question 10 is 4 marks)

A graph passes through $A = (2, 5)$ and $B = (-4, -1)$. A transformation maps every point (x, y) to $(x + 3, 2y)$.

(a) Find the images of A and B. (2)

(b) The original graph is $y = f(x)$. Write the equation of its image and state the original coordinates of a point whose image is $(7, 8)$. (3)

(Total for Question 11 is 5 marks)

C is the curve $y = (x + 1)^3$.

(a) State the coordinates where C meets the coordinate axes. (2)

(b) C is reflected in the y-axis. Find

(i) the equation of the reflected curve, (2)

(ii) the image of each intercept from part (a). (2)

(Total for Question 12 is 6 marks)

13.

The graph of $y = f(x)$ has a maximum point $(2, 6)$ and a minimum point $(5, -3)$.

(a) Find the maximum and minimum points of $y = f(x) + 4$. (2)

(b) Find the maximum and minimum points of $y = -2f(x)$. (3)

(Total for Question 13 is 5 marks)

A curve is obtained from $y = 1/x$ by a translation. Its asymptotes are $x = -4$ and $y = 2$.

(a) Write the equation of the curve. (2)

(b) Find its exact intersections with the coordinate axes. (3)

(Total for Question 14 is 5 marks)

$f(x) = x(x - 3)(x + 2)$.

(a) Find the roots of $f(x) = 0$ and sketch $y = f(x)$. (3)

(b) A second curve has equation $y = f(x + 1)$. Find its roots and the coordinates of the point corresponding to $(0, 0)$. (3)

(Total for Question 15 is 6 marks)

The graph of $y = f(x)$ contains the points $(-2, 7)$ and $(3, -1)$.

(a) For the graph $y = f(2x - 4)$, find

(i) the corresponding two points, (2)

(ii) the point on $y = f(x)$ corresponding to $(5, 9)$ on the new graph. (2)

(b) Describe a sequence of two transformations taking $y = f(x)$ to $y = f(2x - 4)$. (2)

(Total for Question 16 is 6 marks)

C is the curve $y = \dfrac{2}{x^2} + 1$.

(a) State its asymptotes and explain why it has no x-intercepts. (3)

(b) Find the coordinates where C meets the horizontal line $y = 9$. (2)

(Total for Question 17 is 5 marks)

A point $P = (a, b)$ on $y = f(x)$ is mapped to $(4, -6)$ on $y = -3f(x + 2)$.

(a) Find a and b. (2)

(b) Find the image of P on $y = f(x) - 5$. (2)

(Total for Question 18 is 4 marks)

19.

A parabola has equation $y = (x - 4)^2 - 9$.

(a) Find its minimum point and its x-intercepts. (3)

(b) The parabola is translated so that its minimum point moves to the origin. Find the translation vector and the equation of the resulting curve. (2)

(Total for Question 19 is 5 marks)

The curve C is $y = x^3 - 8$.

(a) Find the coordinates of its intersections with the axes. (2)

(b) Curve D is obtained by reflecting C in the x-axis. Find the equation of D and state where D meets C. (3)

(Total for Question 20 is 5 marks)

Figure 21 shows $y = f(x)$ with marked points $A = (-3, -1)$, $B = (0, 2)$ and $C = (4, 0)$. Straight segments join A to B and B to C.

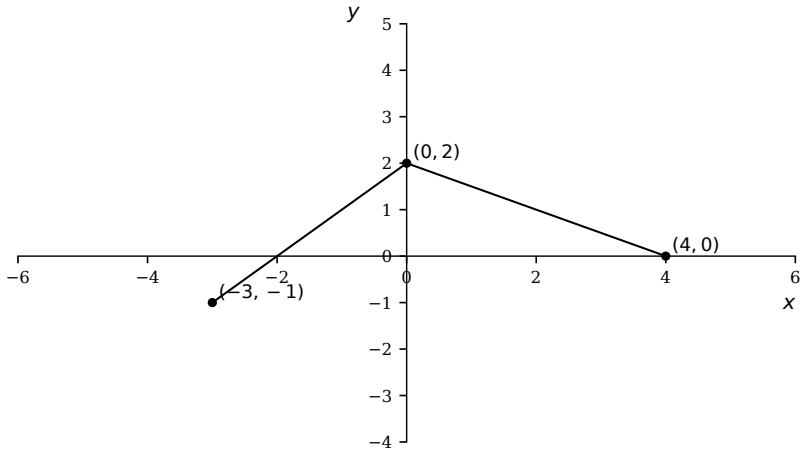


Figure 21

(a) Sketch $y = 2f(x)$, marking the three corresponding points. (3)

(b) Sketch $y = f(2x)$ on separate axes, marking the three corresponding points. (3)

(Total for Question 21 is 6 marks)

The curve $y = f(x)$ is reflected in the y-axis and then translated right by 3.

(a) Write the equation of the resulting curve in terms of f. (3)

(b) If the order of these transformations is reversed, find the resulting equation and the image of the point $(2, 5)$. (3)

(Total for Question 22 is 6 marks)

The curve C is $y = \dfrac{a}{x-1} + 2$, where a is a non-zero constant. It passes through $(3, 5)$.

(a) Find a. (2)

(b) For this value of a, find

(i) the intersections of C with the axes, (2)

(ii) the translation that maps C onto $y = 6/x$. (2)

(Total for Question 23 is 6 marks)

$f(x) = (x - 1)(x + 2)^2$.

(a) Sketch $y = f(x)$, showing all axis intersections and distinguishing a crossing from a touching point. (3)

(b) For the graph $y = f(x) - f(0)$, find the y-intercept and state the vertical translation from the original graph. (2)

(Total for Question 24 is 5 marks)

The curve C has equation $y = \dfrac{3}{(x+2)^2}$.

(a) State its asymptotes and its intersection with the y-axis. (2)

(b) The curve is translated down by 3. Find the equation of its image and all x-intercepts. (3)

(Total for Question 25 is 5 marks)

A curve $y = f(x)$ passes through $(2, 0)$ and $(0, -6)$. Its image under a translation is $y = f(x - 3) + 4$.

(a) Find the images of the two given points. (2)

(b) Explain why the image of $(2, 0)$ need not be an x-intercept of the new curve, and state the equation that determines its actual x-intercepts. (2)

(Total for Question 26 is 4 marks)

The curve $y = f(x)$ is stretched parallel to the x-axis by factor 3 and then reflected in the x-axis.

(a) Write the equation of the resulting curve. (2)

(b) Point (a, b) on the original curve maps to $(-6, 5)$. Find

(i) a and b, (2)

(ii) the image of (a, b) under translation by $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. (2)

(Total for Question 27 is 6 marks)

The curve C is $y = x^2$ and the curve D is $y = (x - 2)^2 + 4$.

(a) Describe the single translation taking C onto D. (2)

(b) Find the coordinates where C and D intersect, and explain why there is only one intersection. (3)

(Total for Question 28 is 5 marks)

Figure 29 shows the graph of $y = f(x)$. It consists of straight line segments joining $A(-4, 1)$, $B(-1, -2)$ and $C(3, 2)$, in that order.

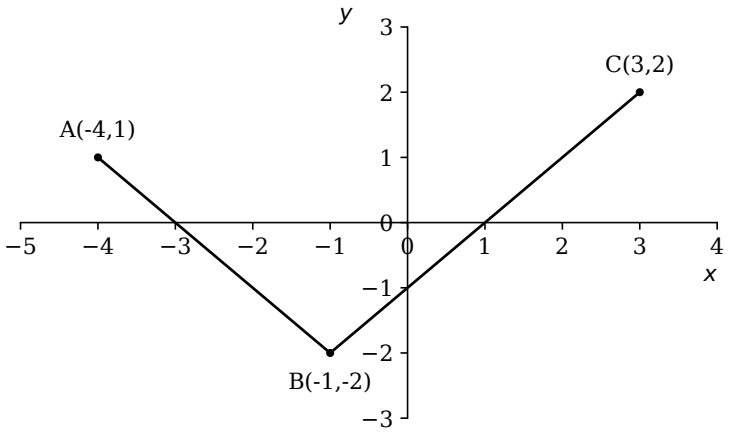


Figure 29

(a) Sketch the graph of $y = -2f(x + 1)$, giving the coordinates of the images of A, B, C. (3)

(b) The graph of $y = f(x)$ crosses the x-axis at two points.

(i) Find their coordinates. (2)

(ii) Hence solve $f(x + 1) = 0$. (1)

(Total for Question 29 is 6 marks)

The curve C has equation $y = x(x - 2)(x + 3)$.

(a) Sketch C, showing all axis intercepts and the behaviour of the curve for large positive and large negative x. Do not calculate turning points. (3)

(b) The curve is translated 2 units to the right to form D.

(i) Find an equation for D in factorised form. (2)

(ii) Find the interval on which D lies below the x-axis and has positive x. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C04-R01 · 6 marks

(a) · 3 marks

x-intercepts $(-2,0),(1,0),(4,0)$; y-intercept $(0,8)$; cubic rises to the right.

三个不同的一次因子给三个穿越点；代入0得8；首项x^3决定左下右上。

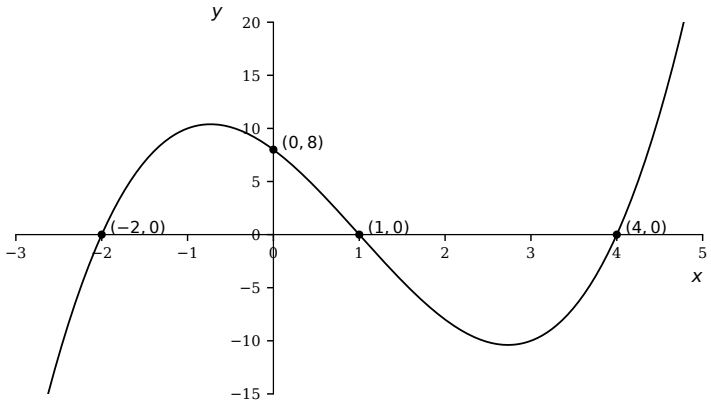
M1	三个横截距
B1	纵截距
A1	正确三次曲线走向

(b) · 3 marks

Image $(3,6)$; $y=(x-1)(x-4)(x-7)-2$.

点坐标加平移向量；方程以x−3替换x再减2。

B1	平移后坐标
M1	正确替换自变量
A1	完整方程



Suggested sketch

Question 2

P1-C04-R02 · 7 marks

(a) For the graph $y = 2f(x-1)+3$, find the images of

(i) · 2 marks

$(-2, 7)$.

x坐标加1，y坐标乘2再加3。

M1	正确变换横纵坐标
A1	点坐标

(ii) · 2 marks

$(2, -5)$ and $(5, 5)$.

两点都应用映射$(x, y) \mapsto (x+1, 2y+3)$，顺序保持。

M1	两点使用一致映射
A1	两个像点

(b) · 3 marks

Translate right 1, stretch parallel to the y-axis by factor 2, then translate up 3.

横移只改x位置，纵向伸缩先于向上平移；如先上移3再伸缩会误成上移6。

M1	正确横移
M1	正确纵向伸缩
A1	上移和顺序

Question 3

P1-C04-R03 · 7 marks

(a) · 3 marks

Asymptotes $x = 0, y = 0$; points $(2, 2), (-2, -2)$; branches in quadrants I and III.

正系数倒数曲线在第一、第三象限，不能画过渐近线；直接代入±2。

B1	两渐近线
B1	两点
A1	两支走向

(b) The curve is translated 3 units left and 1 unit down. Find

(i) · 2 marks

$y = \dfrac{4}{x + 3} - 1.$

向左3用x+3替换，再减1。

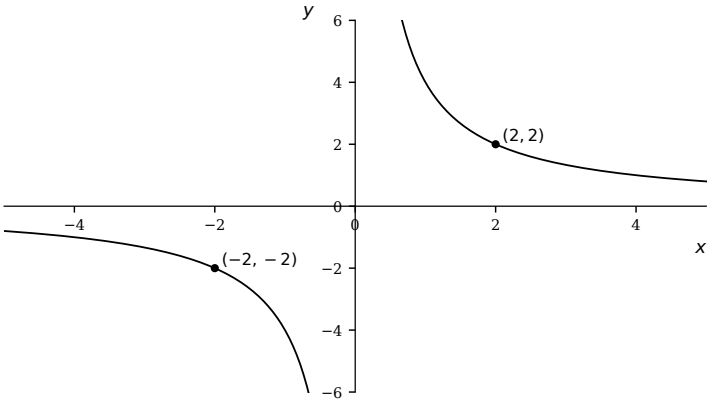
M1	正确水平替换
A1	完整方程

(ii) · 2 marks

$(1, 0)$ and $(0, \frac{1}{3})$.

y=0时x+3=4；x=0时y=4/3−1。

M1	分别令x或y为零
A1	两个截距



Suggested sketch

Question 4

P1-C04-R04 · 5 marks

(a) · 2 marks

$(-4, 0), (-1, -3), (2, 2), (5, 0)$.

关于x轴反射，横坐标不变，纵坐标变号。

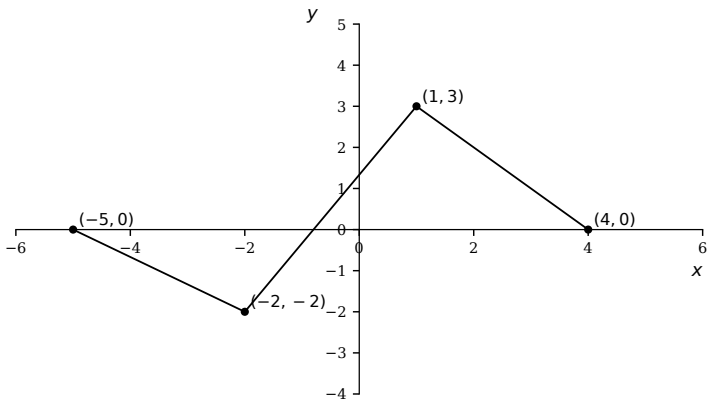
M1	正确映射规律
A1	四点正确

(b) · 3 marks

$(-5, 0), (-2, -2), (1, 3), (4, 0)$, joined in this left-to-right order.

关于y轴反射，横坐标变号，并重新按从左到右顺序连线。

M1	四横坐标反号
A1	四像点
A1	正确连接顺序



Suggested sketch

Question 5

P1-C04-R05 · 8 marks

(a) · 3 marks

Minimum $(2, 1)$; $y \geq 1 > 0$.

平方项非负，最低纵坐标1，故没有横截距。

B1	顶点
M1	用平方非负
A1	无交点理由

(b) C is stretched parallel to the y-axis by factor 3, then translated down 5. Find

(i) · 3 marks

$y = 3(x-2)^2 - 2$; minimum $(2, -2)$.

原整体y值乘3后减5，常数3−5=−2。

M1	变换整个原式
A1	方程
B1	顶点

(ii) · 2 marks

$x = 2 \pm \dfrac{\sqrt{6}}{3}$.

令y=0得$(x-2)^2=2/3$。

M1	解平方方程
A1	两精确根

Question 6

P1-C04-R06 · 6 marks

(a) · 3 marks

$x = 0, y = 0$; $(-1, 6), (1, 6)$; two positive branches symmetric about the y-axis.

x^2正使两支均在x轴上方，且左右对称；趋近x=0时向上无界。

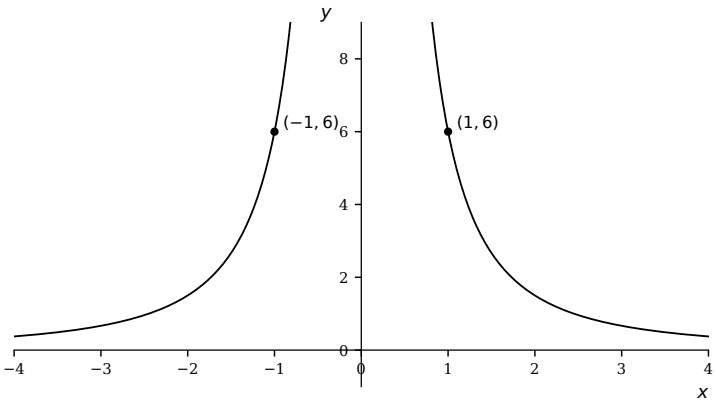
B1	渐近线
B1	两点
A1	正确形状与对称

(b) · 3 marks

$y = 2 - 6/x^2$; $(\pm\sqrt{3}, 0)$.

先反射变为$-6/x^2$，再上移2；令其为0解$x^2=3$。

M1	正确组合变换
A1	方程
A1	两截距



Suggested sketch

Question 7

P1-C04-R07 · 6 marks

(a) Find the corresponding point on each graph:

(i) · 2 marks

$(3, -3)$.

要使函数输入仍为6，需2x=6；y不变。

M1	正确处理水平因子
A1	点坐标

(ii) · 2 marks

$(6, -\frac{3}{2})$.

x不变，y乘1/2。

M1	纵坐标减半
A1	点坐标

(b) · 2 marks

$(\frac{a}{2}, \frac{b}{2})$.

两个独立映射分别使横纵坐标减半。

M1	正确横坐标
A1	正确纵坐标

Question 8

P1-C04-R08 · 5 marks

(a) · 2 marks

$y = (x-2)^3 - 5$.

平移向量$(2,-5)$，在函数中用$x-2$替换并减5。

M1	正确水平平移
A1	完整式

(b) · 3 marks

$(2+\sqrt[3]{5}, 0)$ and $(0, -13)$.

令$y=0$得到$(x-2)^3=5$；令$x=0$得到$-8-5$。

M1	解立方关系
A1	横截距
A1	纵截距

Question 9

P1-C04-R09 · 5 marks

(a) · 2 marks

$(-\frac{2}{3},0),(\frac{1}{3},0),(\frac{5}{3},0)$.

输入3x分别等于原三个零点，所以每个横坐标除以3。

M1	正确水平缩放
A1	全部截距

(b) · 3 marks

$(2,0),(5,0),(9,0)$; corresponding point $(4,4)$.

所有点横坐标加4，纵坐标保持。

M1	统一横移
A1	三个截距
B1	对应点

Question 10

P1-C04-R10 · 4 marks

(a) · 2 marks

A: $x = 2, y = 3$; B: $x = -1, y = -2$.

分母为零给竖直渐近线，外加常数给水平渐近线。

B1	A的两条渐近线
B1	B的两条渐近线

(b) · 2 marks

Translation by $\begin{pmatrix} -3 \\ -5 \end{pmatrix}$.

竖直渐近线从2移到-1，水平线从3移到-2；同样移动整个曲线。

M1	正确水平位移
A1	完整向量

Question 11

P1-C04-R11 · 5 marks

(a) · 2 marks

$A \mapsto (5, 10)$; $B \mapsto (-1, -2)$.

逐坐标应用给定映射。

M1	对两点使用映射
A1	两像点

(b) · 3 marks

$y = 2f(x - 3)$; original point $(4, 4)$.

方程水平位移用x−3；逆推点坐标时横坐标减3，纵坐标除2。

M1	方程水平处理正确
A1	完整方程
A1	逆推点

Question 12

P1-C04-R12 · 6 marks

(a) · 2 marks

$(-1, 0)$ and $(0, 1)$.

分别令y=0和x=0。

B1	横截距
B1	纵截距

(b) C is reflected in the y-axis. Find

(i) · 2 marks

$y = (1 - x)^3$.

用−x替换原式中的x。

M1	正确反射替换
A1	方程

(ii) · 2 marks

$(1, 0)$ and $(0, 1)$.

横坐标反号；纵轴上的点不动。

M1	正确坐标规律
A1	两像点

Question 13

P1-C04-R13 · 5 marks

(a) · 2 marks

Maximum $(2, 10)$; minimum $(5, 1)$.

纵向平移不改变横坐标和极值类型。

M1	两纵坐标加4
A1	两极值点

(b) · 3 marks

Maximum $(5, 6)$; minimum $(2, -12)$.

纵坐标乘-2，负因子会反射并互换最大和最小类型。

M1	坐标乘负因子
M1	交换极值类型
A1	两个正确分类的点

Question 14

P1-C04-R14 · 5 marks

(a) · 2 marks

$y = \dfrac{1}{x+4} + 2.$

水平平移−4、竖直平移2。

M1	正确平移参数
A1	方程

(b) · 3 marks

$(-\frac{9}{2}, 0)$ and $(0, \frac{9}{4})$.

横截距解1/(x+4)=−2；纵截距代入0。

M1	分别设轴条件
A1	横截距
A1	纵截距

Question 15

P1-C04-R15 · 6 marks

(a) · 3 marks

Roots $-2, 0, 3$; positive-leading cubic crossing all three.

每个一次因子各给一零点，首项系数为正；原点同时为纵截距。

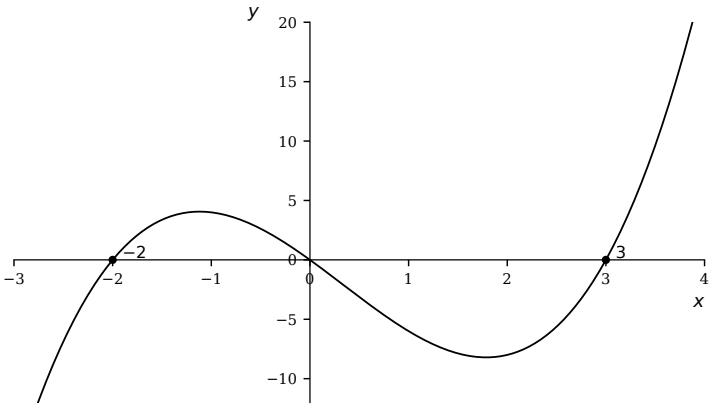
B1	三根
M1	三个穿越点
A1	正确端部走向

(b) · 3 marks

Roots $-3, -1, 2$; image $(-1, 0)$.

x+1等于原根，所有横坐标减1。

M1	正确水平映射
A1	三根
B1	原点的像



Suggested sketch

Question 16

P1-C04-R16 · 6 marks

(a) For the graph $y = f(2x - 4)$, find

(i) · 2 marks

$(1, 7)$ and $(\frac{7}{2}, -1)$.

新横坐标满足2x−4等于原横坐标，所以x=(原x+4)/2。

M1	正确输入对应关系
A1	两坐标

(ii) · 2 marks

$(6, 9)$.

新x=5时原输入$2\times5-4=6$，纵坐标不变。

M1	正确反向对应
A1	点坐标

(b) · 2 marks

Stretch parallel to the x-axis by factor $1/2$, then translate right 2.

由横坐标映射x旧/2+2判断顺序；先压缩后向右移2。

M1	正确水平因子
A1	平移及顺序

Question 17

P1-C04-R17 · 5 marks

(a) · 3 marks

$x = 0$, $y = 1$; $y > 1$ for all permitted x.

反平方项严格正，所以整条曲线在y=1之上；x=0无定义。

B1	两渐近线
M1	利用正反平方项
A1	无横截距理由

(b) · 2 marks

$(-\frac{1}{2}, 9)$ and $(\frac{1}{2}, 9)$.

$2/x^2=8$，故$x^2=1/4$，保留两个符号。

M1	正确平方关系
A1	两坐标

Question 18

P1-C04-R18 · 4 marks

(a) · 2 marks

$a = 6, b = 2.$

新横坐标a−2=4，纵坐标−3b=−6。

M1	正确两坐标关系
A1	两值

(b) · 2 marks

$(6, -3).$

回到原点P=(6,2)，仅把纵坐标减5。

M1	用原坐标而非已变换坐标
A1	像点

Question 19

P1-C04-R19 · 5 marks

(a) · 3 marks

Minimum $(4,-9)$; intercepts $(1,0),(7,0)$.

顶点由配方式直接读出；令平方等于9求截距。

B1	顶点
M1	解平方方程
A1	两截距

(b) · 2 marks

$\binom{-4}{9}$; $y=x^2$.

把顶点(4,−9)移到(0,0)，故左移4上移9，消去配方中的两平移。

M1	正确平移向量
A1	新方程

Question 20

P1-C04-R20 · 5 marks

(a) · 2 marks

$(2, 0)$ and $(0, -8)$.

令$x^3=8$求横截距；$x=0$得纵截距。

M1	解立方关系
A1	两坐标

(b) · 3 marks

D: $y = 8 - x^3$; they meet at $(2, 0)$.

反射使y变号；交点满足$x^3-8=8-x^3$，得$x^3=8$。

M1	正确反射方程
M1	联立两曲线
A1	交点

Question 21

P1-C04-R21 · 6 marks

(a) · 3 marks

$(-3,-2),(0,4),(4,0)$, joined by straight segments.

纵坐标乘2、横坐标不变，保持线段连接关系。

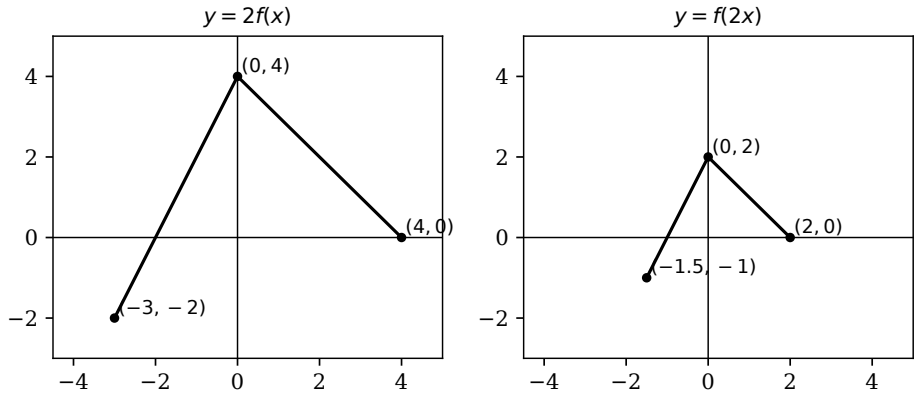
M1	正确纵向伸缩
A1	三点
A1	连线

(b) · 3 marks

$(-\frac{3}{2},-1),(0,2),(2,0)$, joined by straight segments.

横坐标除2、纵坐标不变；与(a)的纵向变换不同。

M1	正确水平伸缩
A1	三点
A1	连线



Suggested sketch

Question 22

P1-C04-R22 · 6 marks

(a) · 3 marks

$y = f(3 - x)$.

反射后为f(−x)，再右移3得f(−(x−3))。

M1	正确反射
M1	对反射结果平移
A1	最终式

(b) · 3 marks

$y = f(-x - 3)$; image $(-5, 5)$.

先右移把(2,5)变(5,5)，再反射成(−5,5)；函数为f(−x−3)。

M1	正确处理逆序函数
A1	方程
A1	点坐标

Question 23

P1-C04-R23 · 6 marks

(a) · 2 marks

$a = 6$.

代入5=a/2+2给a=6。

M1	正确代入已知点
A1	参数

(b) For this value of a, find

(i) · 2 marks

$(-2, 0)$ and $(0, -4)$.

y=0给x−1=−3；x=0给−6+2。

M1	分别设轴条件
A1	两截距

(ii) · 2 marks

$\begin{pmatrix} -1 \\ -2 \end{pmatrix}$.

把渐近线x=1,y=2分别移到坐标轴，需左1下2。

M1	使用两渐近线位移
A1	向量

Question 24

P1-C04-R24 · 5 marks

(a) · 3 marks

Touches at $(-2, 0)$, crosses at $(1, 0)$; y-intercept $(0, -4)$; rises to the right.

平方因子在−2不改变符号，简单因子在1改变符号；首项正，代0得−4。

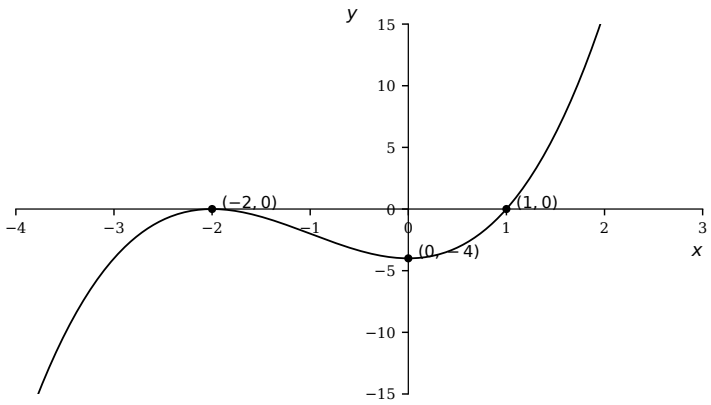
M1	两横截距及不同接触方式
B1	纵截距
A1	整体走向

(b) · 2 marks

Intercept $(0, 0)$; translation up 4.

f(0)=−4，减f(0)即加4。

M1	正确使用f(0)符号
A1	截距与方向



Suggested sketch

Question 25

P1-C04-R25 · 5 marks

(a) · 2 marks

$x=-2, y=0$; $(0, \frac{3}{4})$.

分母为零的竖线是x=−2，反平方远处趋于0；代0求截距。

B1	两渐近线
A1	纵截距

(b) · 3 marks

$y=3/(x+2)^2-3$; $(-3,0), (-1,0)$.

令$3/(x+2)^2=3$得$(x+2)^2=1$。

M1	正确新方程
M1	解平方关系
A1	两截距

Question 26

P1-C04-R26 · 4 marks

(a) · 2 marks

$(5, 4)$ and $(3, -2)$.

点映射为(x+3,y+4)。

M1	正确映射
A1	两点

(b) · 2 marks

The image has $y = 4 \neq 0$; actual intercepts satisfy $f(x - 3) = -4$.

上移会把原横截距离开x轴；新截距应令整个新函数值为0。

M1	用像点纵坐标解释
A1	正确截距方程

Question 27

P1-C04-R27 · 6 marks

(a) · 2 marks

$y = -f(x/3)$.

水平伸长3意味着函数输入除3；再使函数值变号。

M1	正确水平替换
A1	反射符号

(b) Point (a, b) on the original curve maps to $(-6, 5)$. Find

(i) · 2 marks

$a = -2$, $b = -5$.

新坐标为(3a,−b)，与给定像点比较。

M1	正确两坐标关系
A1	a与b

(ii) · 2 marks

$(0, -6)$.

用原点(−2,−5)加向量(2,−1)。

M1	使用原点坐标
A1	新像点

Question 28

P1-C04-R28 · 5 marks

(a) · 2 marks

$\binom{2}{4}$.

配方形式显示顶点由$(0,0)$到$(2,4)$。

M1	正确水平平移
A1	完整向量

(b) · 3 marks

$(2, 4)$; subtraction reduces to $4x = 8$, which has one solution.

联立$x^2=x^2-4x+8$，二次项抵消成一次方程；回代得4。

M1	联立并消去平方项
A1	交点
B1	只一个解的理由

Question 29

P1-C04-R29 · 6 marks

(a) · 3 marks

$(-5,-2),(-2,4),(2,-4)$.

输入$x+1$使横坐标减1；输出乘-2使纵坐标乘-2，按原顺序连接。

M1	水平平移正确
M1	纵向反射伸缩正确
A1	三点与连接

(b) The graph of $y=f(x)$ crosses the x-axis at two points.

(i) · 2 marks

$(-3,0),(1,0)$.

AB直线为$y=-x-3$，BC为$y=x-1$，分别令$y=0$。

M1	利用线段求零点
A1	两零点

(ii) · 1 marks

$x=-4,0$.

输入$x+1=-3$或1，解得$x=-4,0$。

B1	变换后零点

Question 30

P1-C04-R30 · 7 marks

(a) · 3 marks

Zeros $-3, 0, 2$; left end down, right end up.

三个一次因式均为单根，曲线在$-3, 0, 2$穿过x轴。首项x^3决定左下右上；无需超出题目要求求转折点。

B1	三个截距
B1	端点趋势
B1	连贯三次图像

(b) The curve is translated 2 units to the right to form D.

(i) · 2 marks

$y = (x-2)(x-4)(x+1)$.

所有x用$x-2$代入，保留因式结构。

M1	正确输入替换
A1	因式形式

(ii) · 2 marks

$2 < x < 4$.

D零点为$-1, 2, 4$；结合三次曲线符号和$x > 0$得到唯一所求区间。

M1	图像符号
A1	与正横坐标取交集